



Social Accounts for the ocean: A path to inclusive and equitable ocean decision-making

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ABSTRACT

Progress towards equitable ocean sustainability will only be achieved when decision-makers engage with, invest in and use social data. This information is required for decision-makers to identify trade-offs and leverage points to make informed, just, purposeful and defensible decisions that bolster rather than undermine a sustainable

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ocean economy and human well-being. The Ocean Accounts Framework presents an opportunity to integrate social, cultural and equity data in ocean decision-making. Ocean Accounts organize economic, environmental and social information to enable coherent measurement of progress towards the sustainable development of the ocean. In this article, we draw on the collective views of the Global Ocean Accounts Partnership Social Accounts Working Group to explore (i) how Social Accounts can be conceptualized, (ii) an action agenda for progressing the development and integration of Social Accounts into ocean accounting and decision-making and (iii) key principles that should be considered during the development of Social Accounts (including strategic partnerships, stakeholder alignment, inclusive design, transparency and respect for rights). We provide concrete and ambitious actions, which include: (i) developing a flexible and adaptable Social Accounts definition, (ii) undertaking pilot studies using existing data, (iii) developing a standardized structure for Social Accounts and (iv) communicating and advocating for the use of Social Accounts globally. Progress on Social Accounts has the potential to increase the visibility of communities and human-ocean relationships in decision-making and may aid progress towards achieving international commitments.

1. Introduction

Ocean Accounts provide a structured compilation of consistent and comparable information and measure progress across environmental, social and economic domains at regular accounting intervals (Global Ocean Accounts Partnership, 2021). This includes compilation of maps, data, statistics and indicators. The primary purpose of these accounts is to inform and guide public policy and decision-making related to marine environments and resource uses and to support related analysis and research. Ocean Accounts are aligned with relevant international statistical standards such as the System of National Accounts (SNA), System of Environmental Economic Accounting (SEEA) and the Framework for the Development of Environment Statistics (FDES). Hence, the Ocean Accounts Framework can provide a far more holistic understanding of the contribution of oceans to societal well-being than current economic and nature metrics alone (Findlay et al., 2020).

A key component of the Ocean Accounts Framework is the integration of social information (i.e. "Social Accounts"; Global Ocean Accounts Partnership, 2021)^{1,2}. However, there has been limited work in the academic literature that examines Social Accounts and related antecedents within this context (e.g. Loureiro et al., 2023; Perkiss et al., 2022). Hence, the social, cultural, and equity components and contexts of oceans and coastal communities have remained insufficiently represented. Integrating Social Accounts data into ocean accounting provides an opportunity to give policymakers the best available evidence and enhance transparency and accountability within decision-making (Shellock and James, 2024). It would also enable more equitable and culturally sensitive decision-making that ensures that benefits derived from the oceans are distributed fairly, especially for those who have been historically excluded from such access and decision-making processes. Moreover, the 2030 Agenda for Sustainable Development (including SDG14, 15, and 17), already recognizes the importance of disaggregating data to highlight various segments of the population, including women, Indigenous Peoples, and other marginalized groups (Global Ocean Accounts Partnership, 2021). This is essential for meeting broader international commitments and initiatives including the (i) Global Biodiversity Framework targets (e.g. target 14; ensure the full integration of biodiversity) and (ii) the High-Level Panel for a

Sustainable Ocean Economy. Social Accounts play a key role in these efforts by creating the enabling environment necessary for informed and equitable decision-making and producing standard indicators to track progress.

This perspective piece provides the first deep dive into "Social Accounts" in the context of the ocean. Thus far, there has been limited focus on how to conceptualize Social Accounts within the Ocean Accounts Framework and how to develop them in practice. We integrated the collective experiences and perspectives from two working group meetings organized by the Global Ocean Accounts Partnership (GOAP) at the 5th Global Ocean Dialogue in Bali on the 3rd and 4th of July 2024. This paper could serve as a resource for a wide range of global stakeholders, including researchers, ocean accounting practitioners and ocean decision-makers. We believe this could particularly benefit ministries and statistical agencies that participate in, contribute to, or use Ocean Accounts and seek to integrate social dimensions into their frameworks. It may also be of use to stakeholders who are new to ocean accounting and are interested in how the ocean accounting framework can be used to input social, cultural and equity data into decision-making. We hope that our reflections will guide the development and implementation of Social Accounts globally and, help to mobilize countries and stakeholders who want to be champions and contribute to the thought leadership for Social Accounts.

2. Methods

2.1. Approach

To collect views from a diverse group of ocean accounting practitioners, two 1.5 h workshops were held at the 5th Global Ocean Dialogue in Bali on the 3rd and 4th of July 2024. The workshops began with an introductory presentation providing background context and guidance for the workshop discussions. This was followed by facilitated discussion where participants were split into breakout groups and they explored 4 key questions across the two workshops:

- (i) What should be captured in Social Accounts?
- (ii) What types of frameworks and approaches could be used to support these efforts?
- (iii) What actions could be implemented to advance social accounting²?
- (iv) How would you prioritize these actions?

Results from the breakout groups were collected through note taking by workshop participants and the workshop facilitators (RJS and RET). These notes were then transcribed, with the most common themes identified through a manual coding process conducted by the lead authors (RJS and RET) following standard qualitative data analysis techniques (Braun and Clarke, 2006). A thematic coding process was used to identify similar topics and dominant themes. The lead authors synthesized the analysis (RJS and RET) and shared it with co-authors

¹ Our conceptualization of Social Accounts represents a departure from the traditional conceptualization of social accounting found in the established literature, which has developed over more than three decades primarily within corporate and organizational contexts. The shared nomenclature reflects coincidental linguistic similarity rather than any conceptual or methodological relationship. See Section 3.2 for more information.

² "Social accounting" refers to the overarching process of identifying, collecting, and organizing data on the social, cultural, and equity dimensions of human-ocean interactions. It includes the methodologies and tools used to measure and evaluate these dimensions. In contrast, "Social Accounts" are the structured outputs of this process that present the findings in a format suitable for decision-making, monitoring, and advocacy.

responsible for drafting specific sections of the paper (RJS, RET, PASJ, TGL, CR, MS, EK, OL, CC, RA, KLL, EAB, AR, GF). The paper was then written up, giving all co-authors the opportunity to clarify, corroborate, or approve the findings (Bryman, 2012; Shellock et al., 2022). Through this collective reflection, the relevance and value of the identified themes were further validated. Additionally, it provided an opportunity for co-authors to introduce new or emerging themes that extended beyond those discussed during the workshop. The action agenda and principles were further informed and supported by insights from the peer-reviewed and grey literature. For example, from fields such as ocean accounting, Natural Capital Accounting, Marine Social Science, Environmental Science, Marine Science, Development Economics, international development, environmental governance and sustainability governance.

2.2. Researcher positionality

It is best practice for social science scholars to examine how their own positioning might contribute to their interpretations of people's lived experiences (i.e., researchers are not separate from the social processes they study; Kaikkonen et al., 2024). As a group, we are reflexive on the ways in which our personal characteristics (e.g., gender, ethnicity, age, nationality, sexual orientation, immigration status, personal experiences, linguistic tradition, theoretical, political, and ideological stances; Holmes, 2020) shape our perceptions of how to integrate social information into the Ocean Accounts framework. This is particularly important due to the dimension of "equity" which is central to Social Accounts. Hence, we have provided a positionality and reflexive statement (Moon and Blackman, 2014; Secules et al., 2021). Our work is both constrained and charged by our positionality.

In total, 33 authors contributed to this perspective piece. The author team was composed of 21 women and 12 men. We write from the position of being non-Indigenous scholars, which is indicative of structural inequities where we work (Weir et al., 2024). The authors are based in 11 different countries: (i) Indonesia (n = 8), (ii) Australia (n = 7), (iii) UK (n = 4), (iv) Kenya (n = 2), (v) USA (n = 2), (vi) The Netherlands (n = 2), (vii) Norway (n = 2), (viii) Thailand (n = 1), (ix) Antigua and Barbuda (n = 1), (x) Spain (n = 1) and (xi) Mozambique (n = 1). Fifteen of the authors reside in a different country from their birthplace. For instance, UK, Brazil, China, Ecuador, France, Fiji, Iraq, and Italy. This perspective piece features contributions from Indonesia and other Global Majority contexts (n = 12), highlighting voices often marginalized in the ocean governance literature.

Most authors worked in education and academic research (n = 12) or non-governmental organizations (n = 9). The authors also worked in consultancy research (n = 4), government and policy (e.g. Ministries, Statistical Agencies and Arm's Length Bodies; n = 5), funding bodies (n = 1), or were currently unemployed (n = 1). This creates a blend of research, advocacy, and policy implementation perspectives. However, this sector-positioning could bias the perspective piece towards research-driven and policy-reform solutions rather than industry or grassroots perspectives. This could have inadvertently privileged western science or academic ways of knowing over other perspectives and knowledge systems. In terms of their position, most authors were mid-level professionals (n = 9) and senior professionals or experts (n = 10). The remaining were in: (i) entry-level or junior positions (n = 3), (ii) managers and supervisors (n = 3), (iii) Entrepreneur/Founder (n = 3), (iv) Directors (n = 2), and executive or senior leadership (n = 1). Hence, this perspective piece has been informed by both emerging and established career experiences.

Authors came from a range of different natural science and social science disciplinary backgrounds. 25 natural science disciplines were mentioned (e.g. marine biology, oceanography, marine ecology, chemistry, marine conservation). Authors also originated from 16 different social science disciplines (e.g. Environmental Economics, Marine Social Science, accounting, and development). The authors have a range of

research interests. Examples include Ocean Accounts, Natural Capital, ocean governance, coastal management, monetary valuation, Marine Spatial Planning, ocean justice and equity, blue economy, coastal communities and, sustainable fisheries. Our disciplinary training, spanning natural sciences and social sciences, likely influences what we consider valid forms of knowledge and how we conceptualize the relationship between social and ecological systems. The interdisciplinary nature of the co-author group positions us to see connections and tensions that single-discipline approaches might miss, particularly around the complex socio-ecological challenges facing ocean systems. We approach ocean governance from positions that generally assume: (1) scientific knowledge should inform policy decisions, (2) current governance systems require significant reform rather than replacement, (3) interdisciplinary collaboration can address complex ocean challenges, (4) equity and justice concerns are central to sustainable ocean futures, and (5) a system thinking-and-design approach is needed to foster necessary changes in governance and management of ocean resources. These beliefs reflect our particular perspective and may differ from how other groups, sectors, communities, or governments view ocean challenges.

3. Conceptualization of Social Accounts for the Ocean Accounts Framework

3.1. What are Social Accounts within the Ocean Accounts Framework?

Within the Ocean Accounts Framework, Social Accounts provide insights into the societal dimensions of peoples' interactions with ecosystems, including impacts and dependencies. Initial definitions from ocean accounting define Social Accounts as the process of compiling, evaluating, and reporting data across inclusivity, equity, social, and cultural dimensions of the human-ocean relationship (Loureiro et al., 2023; Global Ocean Accounts Partnership, 2021; Holloway et al., 2024). Aligned with the UN SEEA approach to natural capital accounting, this framework is rooted in stocks (i.e., resources at a given time) and flows (i.e., changes in stocks over time), of services from natural capital. Natural Capital Accounting is an approach that integrates environmental and economic information to track how nature supports the economy and how human activity affects the environment. Recent developments in ocean accounting have intentionally diverged from finance- and economic-based notions of social accounting to assess inclusivity, equity, and social impacts of ocean natural capital availability. This SEEA-aligned accounting conceptualization determines that the stocks of natural capital provide flows of social benefits, with Social Accounts helping to illuminate how these flows support social capital and equity. Social Accounts can help to:

- **Evaluate human dependence on ocean services** for well-being, including livelihoods, income, food security, and basic needs.
- **Assess the distribution of benefits and costs of ocean use** across social groups and regions, particularly focusing on Indigenous Peoples and local communities.
- **Identify ocean-related risks** due to anthropogenic activities or environmental changes and evaluate the socio-ecological resilience of affected communities.
- **Consider social conditions and societal assets** occurring at specific points in time, which both shape and are shaped by ocean health and the ocean economy. These include demographic trends, socio-economic conditions, health, education, skills, knowledge, culture, and equity.
- **Understand and increase the visibility of social activities** that may fall outside the traditional production boundary of national accounts (e.g. unpaid household labour and customary marine stewardship activities).
- **Assess the impact of governance and management decisions** on social equity, cultural practices, and community resilience, helping

to understand how policy choices influence ocean-dependent communities.

Within the Ocean Accounts Framework, Social Accounts provide insights into the societal dimensions of peoples' interactions with ecosystems, both "in terms of" impacts and dependencies? They complement and expand the usefulness of measuring ecosystem services, which serve as a bridge between strictly economic metrics, as defined in the SNA, and broader ecosystem measures captured in the SEEA framework. A key challenge still lies in defining the scope of "social" as distinct from "economic" and deciding how, and if, to integrate socioeconomic information into Social Accounts effectively. Aligning with the SEEA Ecosystem Account and SNA frameworks, these accounts illuminate dependencies and contributions related to natural capital service flows. Importantly, they are not strictly tied to the exchange values of ecosystem services, which are measured solely in monetary terms (Fig. 1). Social Accounts therefore complement environmental-economic accounts by including aspects such as cultural connections and community well-being, which are not always captured through environmental-economic indicators or traditional economic metrics alone. The alignment of Social and Ecosystem Accounts also highlights the relationship between metrics of human-well-being and changes ecosystems and their asset values.

Social Accounts further complement Governance Accounts within the Ocean Accounts Framework, which document the policies, institutional arrangements, and legal structures that influence ecosystem management and societal benefits. As the framework evolves, governance elements will be incorporated directly within Social Accounts. Although it is important that governance aspects do not become the sole focus of Social Accounts. By bridging gaps across ecological, economic, and social domains, Social Accounts enhance the utility of Ocean Accounts for informing inclusive and sustainable decision-making.

We build on these initial definitions and initiatives, highlighting 16 key types of data that will need to be considered within Social Accounts (see Fig. 1):

1. Gender equity and social inclusion
2. Jobs, income and ocean-based labour
3. Access and rights to marine resources and services
4. Human security, safety and, disaster preparedness
5. Marine-dependent livelihoods and resource use
6. Food and nutrition security
7. Access to basic needs and welfare
8. Human health
9. Knowledge and skills
10. Indigenous, traditional, and local knowledge and stewardship
11. Blue economy and sustainable trade
12. Social structures and demographic trends
13. Vulnerability and resilience to environmental change
14. Social cohesion and engagement
15. Cultural, sacred, and personal connections to the ocean
16. Nature-based leisure, recreation and tourism

These 16 types of data represent preliminary thinking drawn from the working group's discussions and will be further refined and validated through the proposed action agenda (Section 4). While Social Accounts for the ocean require a distinct perspective to account for the unique characteristics of marine environments—such as fluid boundaries, three-dimensional resource use, and maritime governance structures—the broader applicability of the domains highlights two important opportunities. First, to draw on existing experience and frameworks from social accounting in other ecosystems and adapt or extend them. Second, to transfer and apply advancements to terrestrial ecosystems like forests, mountains, or grasslands with appropriate modifications.

The development of Social Accounts will likely require extending

and going beyond the SEEA and SNA, for example, through the use of hybrid Accounts. This is to ensure that Social Accounts are more holistic and incorporate values that are important in supporting communities and go beyond just monetization.

3.2. How does our framework differ from existing social accounting conceptualizations?

To clarify our framework's distinct position, it is essential to explicitly differentiate our Social Accounts approach from the two predominant interpretations of social accounting found in existing literature. While we share the same terminology, this reflects coincidental linguistic similarity rather than any conceptual relationship. The first is the conceptualisation of "social accounting" which has developed over more than three decades, primarily within corporate and organizational contexts (Gray, 2001; Gray et al., 1996; Mathews, 1997). It aims to address the social impact of economic actions beyond traditional financial metrics (Mook, 2013). This involves reporting on non-financial information to stakeholders to emphasize accountability, social responsibility, and transparency (Retolaza and San-Jose, 2021). The second is the Social Accounting Matrix (SAM), which examines economic transactions within the economy and is derived from the SNA. It is a particular representation of the macro and meso economic accounts of a socio-economic system, which capture the transactions and transfers between all economic agents in the system. They are often used in development planning since they provide additional detail about the relationships between sectors of the economy, social groups, and economic agents, particularly in terms of how income is generated, distributed, and spent. Although it is mentioned in the technical guidance for ocean accounting (Global Ocean Accounts Partnership, 2024, 2021), it falls outside of the scope of our definition of social accounting. SAMs fail to capture the social, cultural and equity factors associated with the ocean. They only deal with socio-economic indicators, such as extending the Input-Output (IO) Tables to include income distribution and institutional transactions.

4. Action agenda for Social Accounts

Drawing on discussions during the working group meetings and the perspectives and experiences of the author group, we have developed an action agenda for Social Accounts. There are eight key areas where efforts should be targeted to progress the development and integration of Social Accounts in ocean accounting and decision-making. They are: (i) establish the state of evidence, (ii) develop a working Social Accounts definition, (iii) start pilot studies and use existing data and indicators, (iv) undertake gap identification, (v) learn from and connect with other disciplines, (vi) develop a standardized structure for Social Accounts, (vii) develop guidance for undertaking Social Accounts and, (viii) communicate and promote Social Accounts. The proposed actions are comprehensive and wide-ranging, enabling implementation by stakeholders across various sectors. Efforts can be led by a single stakeholder, such as a government department, or collaboratively by multiple stakeholders from different sectors, organizations, or departments working together to develop Social Accounts. This is summarized in Fig. 2. Moving forward, we recognize that genuine co-development with communities and stakeholders (e.g. ministries and statistical agencies) is essential. The Action Agenda should be refined through direct consultation with communities and stakeholders to ensure it addresses their actual needs and constraints rather than the authors perceived gaps.

4.1. Establish the state of evidence on Social Accounts

Social accounting is an emerging concept for the ocean that needs to draw on existing knowledge and accounting practices. Examining accounting efforts in terrestrial and freshwater environments can establish whether and how Social Accounts have been conceptualized and

TYPES OF DATA THAT CAN BE INCORPORATED INTO SOCIAL ACCOUNTS



Fig. 1. 16 types of data that can be incorporated into Social Accounts.

implemented in other biomes. It will be important to synthesize this evidence using high-quality and stringent methods (e.g., Systematic Reviews and Systematic Maps) to ensure that researchers, practitioners, and decision-makers have the best available evidence (e.g. Shellock et al., 2024).

4.2. Develop a working definition for Social Accounts

Establishing clear definitions is a critical step in the development of Social Accounts. A working definition of Social Accounts should establish a common language for collaboration and coordination to guide immediate discussions, research, and implementation efforts. A robust and reliable working definition will ensure a consistent, practical approach which meets the statistical requirements of accounting frameworks like SEEA and SNA and enables effective monitoring, reporting, and verification.

However, we also recognize that Social Accounts will evolve as we learn more. There will be a need for a “living definition”, which allows for the dynamic evolution of Social Accounts with advancing knowledge and experience. This living definition can accommodate a growing body of data and insights on human-ocean interactions, allowing the gradual integration of new perspectives and interpretations. Accompanied by the practical, working definition, a living definition ensures that the field remains responsive across geographies, generations, values, knowledge systems, and temporal scales, while maintaining its

fundamental statistical integrity.

4.3. Start pilot studies using existing data and indicators

It is essential to conduct pilot studies to explore how Social Accounts can be delivered in practice. With pilots underway in Costa Rica, Belize, Maldives and Mozambique (<https://www.oceanaccounts.org/social-accounts/>) (see Box 1 for Mozambique Pilot), additional pilot studies should be undertaken in a selection of countries which meet one or many of the criteria (see Supplementary Materials 1), including where there is expertise, partners, capacity, and political will for ocean accounting. Pilot Social Accounts need to be delivered at one of the following scales, depending on the evidence and policy needs (Shellock and James, 2024):

- **Macro-level** (i.e. Social Account at a larger scale, including regional, national, or global populations)
- **Micro-level** (i.e. Social Account focusing on individuals, households, sectors, communities, organizations).

We recognize the need for Social Accounts to draw on diverse datasets, including primary, secondary, formal, and informal sources. However, to enable consistent implementation at scale and at a relatively low cost, initial pilot approaches should be linked to existing, formalized data collection methodologies where possible. Data collected



Fig. 2. Summary of actions proposed for the development and implementation of Social Accounts.

Box 1

Social Accounts pilot in Mozambique.

In Inhambane Bay, Mozambique, the GOAP African Community of Practice (ACoP) is piloting a community-level Social Account (World Resources Institute, 2025). Social Account that draws on both formal secondary data and local knowledge. As part of this effort, which is led by Ocean Revolution Moçambique, the World Resources Institute, and AfriSeas, the team reviewed existing secondary data sources, including national household and fisheries surveys, and identified gaps in thematic coverage and representation. They found limited information on artisanal fishing practices, especially those led by women such as shellfish collecting. The Pilot also revealed broader absences in data on cultural values. For example, the role of marine resources in shaping identity and ceremonial practices and on local and traditional knowledge and marine stewardship more generally. The ongoing pilot is now exploring how participatory and community-based methods can help fill these critical gaps. This Pilot captured: (i) economic, nutritional, and cultural dependencies, (ii) local knowledge of ecosystem change and the (iii) social conditions that shape local capacity to adapt to those changes.

through standardized national household surveys, often called Household Income and Expenditure Surveys (HIES) or similar names across countries, provide robust and regularly collected data in a transparent and replicable manner. Other sources, such as Fisheries or Agricultural Censuses, can provide additional consistent and relevant information, particularly concerning resource production and consumption.

Initial development of Pilot Social Accounts should focus on analyzing household and census questionnaires, methodologies, summary reports, and metadata. As a first step the focus should be on publicly available information so research can continue at pace without the need to request or collect additional data. Initial evaluations should also examine what spatial information, if any, can be used to link survey data to ecosystems. When the analysis of readily available information is complete, specific data requests can be identified and processed. For more comprehensive Social Accounts, additional and targeted data collection efforts can complement existing national surveys to address specific gaps or local needs. However, initial pilots based on existing secondary data will provide valuable insights and learning opportunities to advance the field, especially where resources and capacity are limited.

4.4. Undertake gap identification

In many contexts, there will not be coverage across all the 16 types of social data, as they are not routinely collected. Where data does exist, there may be gaps in methodology, representation, policy alignment, and the use of social data by decision-makers. To develop and implement effective Social Accounts, the delivery of Pilot Social Accounts should be accompanied by a systematic analysis which identifies and

addresses these gaps. We recommend a stepwise approach comprising nine main steps, which are detailed further in Table 1. This stepwise approach helps to identify where gaps exist and the ways in which to fill those gaps. Filling data gaps may present significant challenges where resources and capacity are limited. Recommended interventions should prioritize the most critical gaps for policy and planning (Table 1, Step 6) and may involve adjusting national surveys or establishing partnerships and data-sharing agreements with non-governmental organizations, academic institutions, or civil society organizations.

4.5. Learning from and connecting with other fields and disciplines

Filling gaps with Social Accounts requires learning from other fields to build on existing knowledge and avoid duplicating efforts. Insights from established frameworks, theories, and tools across diverse disciplines will ensure Social Accounts develop efficiently and connect to broader global initiatives.

Fields such as economics, public policy and governance, Marine Social Science, sociology, anthropology, political economy, and human geography can provide crucial insights for shaping Social Accounts. Marine Social Science, for example, has explored the connections between humans and marine ecosystems (Bennett, 2019), including aspects related to Social Accounts such as: (i) social justice, welfare, and equity, (ii) coastal communities, (iii) valuing and connecting with marine environments, (iv) maritime industries, and (v) stakeholder engagement and participation (McKinley et al., 2022).

Building on these disciplinary insights, theoretical approaches such as Nature's Contributions to People (Pascual et al., 2017), intersectionality (Crenshaw, 1989) and gender theory (Reeves and Baden,

Table 1
Stepwise approach to gap identification for Pilot Social Accounts.

Stage number	Stage	Description
1	Define key objectives	Outline the primary goals and intended outcomes of Social Accounts (e.g. tracking gender equity and social inclusion and understanding socio-cultural impacts). They can be informed by mapping out the relevant policies that Social Accounts aim to influence. These goals and outcomes form the objectives against which gaps can be measured.
2	Conduct data inventory and needs assessment	Identify what data is currently available, its quality, and how it aligns with Social Accounts objectives. Consider primary and secondary data sources, such as national statistics, community surveys, and environmental monitoring programs. To assess data gaps, it is essential to determine where there are shortages or limitations in existing data.
3	Review methodological and analytical gaps	Review whether current methods are sufficient to capture the complexities of Social Accounts. This includes examining if existing frameworks can adequately capture intangible values and assessing the need for improved tools or technology to effectively capture or analyse Social Account data.
4	Evaluate inclusivity and equity	Identify representational gaps by checking if specific groups (e.g., Indigenous communities, small-scale fishers, marginalized coastal populations) are underrepresented or absent in current data.
5	Examine access and participation	Review how well the data reflects equitable access to ocean resources and participation in decision-making processes. For example, are there gaps in data on who has access to marine resources or involvement in marine governance?
6	Analyse policy and governance gaps	Determine if existing data aligns with current policy objectives, such as the Sustainable Development Goals (SDGs) or national ocean policies. Determine if and how social data is used to inform or achieve these objectives. Identify areas where additional information, awareness, or capacity is needed to support policy goals (e.g. specific data on ocean-related livelihoods or socio-economic vulnerability). Identify the overlapping governance structures of the marine and coastal environment, the information on these within the available data and the need to compile a governance-focused Social Account.
7	Analyse policy monitoring and accountability gaps	Review if there are gaps in tracking the impact of policies and governance frameworks on different social groups. This could include a need for more robust indicators to assess policy effectiveness or to ensure accountability in marine resource management.
8	Engage communities and stakeholders for ground-level insights	Conduct consultations and engage with communities and stakeholders to identify their data needs,

Table 1 (continued)

Stage number	Stage	Description
9	Compile findings into a gap report	Summarize identified gaps in data, methodology, representation, and policy alignment. Recommend interventions to fill critical policy and governance gaps. These may include adjustments to existing national surveys or partnerships and primary data-sharing agreements with non-governmental organizations, academic institutions, or civil society organizations.

2000) offer additional lenses for analyzing human-environment relationships, social identities, and resource access. Structural elements of Social Accounts can draw on models from climate and disaster risk indices (e.g. Climate Vulnerability Index, Sendai Framework), food security metrics (e.g. the Global Food Security Index) and well-being frameworks (e.g. Thriving Places Index and Doughnut Economics; [Centre for Thriving Places, 2024](#); [Raworth, 2018](#); [What Works Well-being, 2023](#)). These indices offer valuable models for integrating social, environmental, and economic dimensions into holistic measurements of societal progress.

Where existing data and methods are insufficient to capture the complexities of Social Accounts, lessons from socioeconomic monitoring initiatives, like the Global Socioeconomic Monitoring Initiative for Coastal Management ([Hoon et al., 2008](#)) can provide systematic frameworks for measuring human-ocean interactions. For capturing nuanced, localized insights, the field can draw on participatory and community-driven approaches. These include participatory mapping ([Burdon et al., 2019](#)), the Community Voice Method and creative methods, including art, storytelling ([Strand et al., 2022](#)), and visual research techniques like photovoice ([Anderson et al., 2023](#)). Such methods can enable direct community and stakeholder engagement, ensuring that outputs are contextually accurate and locally meaningful.

4.6. Develop a standardized structure for Social Accounts

Critical to the development of Social Accounts is the design of a standardized structure (i.e. accounting tables), which maintains flexibility to national, subnational, and local contexts. A consistent and coherent structure ensures data is presented in a format that is accessible to both policymakers and the public, supporting transparency in decision-making processes. Standardized Social Accounts enable users to compare data over time, identify trends, evaluate progress, and analyze regional differences to support informed policies and management. The accounts must also ensure that they are presented in a manner consistent with the environmental-economic accounting basis of the rest of the framework. There may be a need for multiple accounting tables for Social Accounts. This aligns with other accounts, for example, the Tourism Satellite Accounts use 7–10 tables.

The structure should be informed by: (i) established Ocean Account components, (ii) gap-identification (Action 4.4), (iii) insights from other fields (Action 4.5) and (iv) macro- and micro-level Pilot Accounts (Action 4.3). Building on established accounting structures will ensure

Social Accounts are integrated with environmental and economic components. For instance, this approach enables the disaggregation of ocean economy benefits, such as employment, by gender, socioeconomic status, or residence, facilitating analysis of benefit distribution across different groups.

Just as the SEEA and SNA frameworks align through shared data on environmental pressures and economic activities, Social Accounts overlap with both by integrating data on human well-being, cultural values, and governance. These intersections highlight the potential for Social Accounts to provide a unifying lens, particularly in policies that address cross-cutting themes such as prosperity (e.g. through income), equity, sustainability, and ecosystem governance. Establishing clear linkages between these components ensures consistency and coherence across the ocean accounting framework.

The development of these linked structures should leverage the gap analysis and insights from other fields to identify where existing, well-adopted frameworks are adequate and where new structures need to be created. Additionally, conducting pilots and gap analyses across multiple geographies will be essential to ensure a standardized yet flexible structure that can adapt to the socioeconomic, cultural, and ecological contexts and available data within each region. As Social Accounts aim to fill gaps, a key challenge will be developing structures for place-based dimensions, such as Indigenous and local knowledge and cultural identity, that vary across regions. For these dimensions, a standardized framework may not be feasible, especially at the micro level, where cultural and ecological contexts are highly specific. Instead, where accounts seek to capture place-based dimensions, co-design (see [Section 5.2](#)) will be essential to ensure they are meaningfully represented.

4.7. Develop guidance and capacity for compiling and using Social Accounts

Following actions from 4.1 to 4.6, there will be a need to produce draft guidance for compiling and using Social Accounts. Initial guidance should focus on how secondary data can be used to populate Social Accounts. This will draw on lessons learned from the first Pilot Accounts and provide insights for countries and organizations wishing to establish Social Accounts. Two documents should be produced: (i) technical guidance and (ii) a short summary for policymakers to suit different audiences. This information will also need to feed into the GOAP Technical Guidance ([Global Ocean Accounts Partnership, 2021](#)) to ensure integration with the wider ocean accounting framework.

Creating this guidance should be a collaborative and participatory process. Writing workshops and iterative feedback cycles can facilitate the co-production of content by engaging diverse contributors, including members of the GOAP Social Accounts Working Group, technical experts, government agencies involved in the first Pilot Accounts, and grassroots partners (e.g. Local and Indigenous advisory committees and groups; see [Section 5.2](#)). This approach will ensure the resulting guidance reflects a range of perspectives, expertise, and stakeholder priorities, making it both practical and inclusive.

The guidance should help account developers tailor the type and level of information in Social Accounts to meet the needs of the intended users. It should offer clear recommendations on how to select indicators across dimensions, considering the scale, purpose, and context of the account. Guidance should pay particular attention to how Social Accounts can be tailored to reflect local realities, priorities, and complexities. This tailoring is essential to avoid applying Social Accounts in overly rigid or standardized ways that might overlook locally relevant social or cultural considerations. The guidance should also discuss how to identify appropriate secondary data, organize and disaggregate data effectively, and address ethical considerations for storing and sharing sensitive information.

To support effective use, the guidance should include examples of how Social Accounts can inform ongoing policy processes, such as tracking progress on the Sustainable Development Goals or evaluating

the social impacts of marine or coastal policies. Case studies from the Pilot Accounts will demonstrate how these guiding principles can be practically applied, highlighting both the challenges encountered and the strategies used to address them. The guidance should also cover the ongoing maintenance and use of Social Accounts, including how to select and update indicators over time to ensure continued relevance for policy and decision-making. These practical illustrations will ensure guidance is actionable and user-focused. Building capacity of certain groups of stakeholders such as researchers, natural resource statisticians, government and Civil Society Organizations to implement the guidance is indispensable for the adoption and good use of Social Accounts.

As the field advances, future iterations of the guidance can expand to cover more diverse data types and methodologies. These include community-driven and participatory approaches, which are essential for capturing nuanced, localized insights and knowledge. These efforts should be informed by ground-level engagement during the gap analysis process, tools and insights across established fields, and future pilot research. The phased approach ensures that Social Accounts can launch swiftly by leveraging readily available information, while progressively evolving to incorporate more diverse and complex data sources and methodologies over time.

4.8. Communicate and promote Social Accounts

Effective communication and promotion are essential for ensuring that Social Accounts achieve their intended impact. Social Accounts offer a pragmatic approach to deliver a more holistic view on ocean management issues and potential solutions. However, their utility ultimately depends on stakeholders not only understanding their value but also actively engaging with and supporting their development and implementation. Since the integration of social information into Ocean Accounts is relatively new, clear and compelling communication is necessary to introduce the concept to diverse stakeholders, establish their relevance, and encourage adoption. Communication and promotion can ensure Social Accounts are seen as actionable and credible tools for decision making, encouraging their integration into policy frameworks, funding priorities, and governance structures. Additionally, transparent communication builds trust among stakeholders. This is especially relevant for the inclusion of women, local communities, and Indigenous groups who are often overlooked in ocean management discussions, but who's participation and data could serve as the foundation for key components. Stakeholders should be engaged not just as recipients of information but as active participants in co-designing solutions, ensuring their insights and priorities are reflected in the development of the field.

Various stakeholders will experience different benefits and impacts from Social Accounts, making it essential for the communication strategy to identify and engage diverse range of groups, including local communities, civil society, industry, government, and academia. Tailored messaging and strategies should align with the specific priorities and concerns of each stakeholder group. For example, communications and promotion with governments, should highlight how Social Accounts can align with government agendas such as economic planning, social equity and sustainable development. Messaging can focus on how Social Accounts can improve decision-making, evaluate and inform policies, and track progress towards national and international goals and commitments. This can be informed by mapping the potential uses and policy relevance of Social Accounts within the national context. Advocacy can also highlight the advantages of community-based and participatory accounting approaches, which include improving bottom-up representation, reducing conflict, and overcoming resource limitations to fill data gaps. With local communities, communication could highlight how Social Accounts reflect their realities, amplify their voices, and contribute to tangible, local benefits such as better resource management, services, or recognition of rights.

5. Principles for guiding the development of Social Accounts

Drawing on discussions during the working group meetings and the perspectives and experiences of the author group as well as wider environmental management literature, we have identified a set of principles to guide the development of Social Accounts and the broader field: (i) strategic partnerships and stakeholder alignment; (ii) inclusive design and implementation; and (iii) transparency and respect for rights.

5.1. Strategic partnerships and stakeholder alignment

The integration of social information into Ocean Accounts should be guided by strategic partnership to ensure they are practical, impactful, and aligned with global and regional initiatives. Partnerships can enhance the quality, relevance, and adoption of Social Accounts by leveraging diverse expertise, resources, and networks. When partnerships are facilitated across sectors and scales, they can facilitate coordination between different stakeholders and enable long-term sustainability of Social Accounts.

Strategic stakeholder analysis (or mapping) is a critical step in identifying and prioritizing engagement. Stakeholder analysis helps to systematically analyze who may be affected by an issue, intervention, project, process or decision. This approach is key for identifying stakeholders' interest in, influence over and impact on specific issues, interventions, projects, processes and decisions related to Social Accounts (Reed et al., 2025). This can help to ensure that relevant groups are considered and reduces risks of power groups dominating discussions on Social Accounts, marginalizing other groups and voices (Reed et al., 2025). Analysis should include stakeholders such as environmental and finance ministries, statistical agencies, Indigenous groups, and coastal communities.

For example, engaging a county's statistical authority or equivalent institution could enhance the credibility, comprehensiveness, and integration of Social Accounts into national reporting systems. Furthermore, these agencies can serve as custodians of Social Accounts, ensuring their long-term management and use, as well as enabling the incorporation of data from various sources, including NGOs, communities, and other government agencies, into a formalized and cohesive framework. In addition to statistical agencies, partnerships with relevant government ministries, including environmental and finance ministries, can ensure Social Accounts align with existing policy priorities and global agendas, such as the 30 × 30 initiative and national sustainable development plans. Finally, building an alliance of coastal communities and Indigenous groups is essential for ensuring Social Accounts reflect local realities and priorities.

5.2. Inclusive design and implementation

Developing components of Social Accounts which cover place-based dimensions such as cultural identity or local and Indigenous knowledge requires inclusive approaches, particularly when based on primary data. To ensure these dimensions accurately reflect the diverse perspectives and values of coastal communities, they should be co-designed and co-produced with those they seek to represent, including Indigenous Peoples, women, youth, and other structurally marginalized groups. Co-design involves collaboratively shaping the structure, objectives, and methods from the outset, while co-production ensures ongoing stakeholder involvement throughout implementation (Kaikkonen et al., 2024; Strand et al., 2024). Prioritizing the engagement and participation of these groups during design and implementation helps define structures and shape guidance, briefs, and recommendations that are relevant, effective, culturally appropriate, and beneficial to the communities they represent. At the same time, the design and production of Social Accounts should also strategically consider the needs of other stakeholders, such as government bodies, financiers, and operating companies, ensuring they provide value to a broad range of users.

5.3. Transparency and respect for rights

As the Social Accounts field develops, it is essential to uphold ethical standards that ensure transparency, protect rights, and minimize harm. Ethics must be considered through the delivery of the action agenda and beyond. For example, when gathering insights from other fields (Action 4.5), the field should learn from past experiences to prevent harm and ensure alignment with humanitarian standards. During the Pilot Account efforts (Action 4.3), ethical safeguards for handling sensitive and private data should be implemented, and the development of initial guidance (Action 4.7) should establish clear guidelines for handling social information, including the responsible integration of disaggregated data. A major challenge will be balancing policy usefulness and ethical considerations. In some cases, ensuring anonymity, whether due to sparse populations or few businesses, requires data aggregation that can obscure relevant patterns, and limits its value for decision-making.

Upholding ethical integrity through these principles will become even more crucial as more diverse data sources and methodologies are incorporated, particularly community-driven and participatory approaches. As Social Accounts seek to capture ground-level insights and Indigenous, Traditional and Local (ITL) Knowledge, they must adopt well established ethical safeguards and principles, such as those Indigenous data governance frameworks. This includes the CARE Principles (Collective Benefit, Authority to Control, Responsibility, Ethics; Carroll et al., 2020). These principles will ensure that Social Accounts enhance representation while respecting communities' rights to own and control their knowledge and data (Jennings et al., 2023; Sedran-Price, 2024). By embedding strong ethical standards, social safeguards, and privacy protections, Social Accounts can build an inclusive and secure environment that allows coastal communities to participate confidently, while also mitigating potential risks and unintended negative impacts.

6. Conclusion

As we look to the future of ocean accounting and its role in mainstreaming the ocean's contribution to societal well-being, it is crucial for Social Accounts to be part of ongoing efforts. The paper has provided the first detailed conceptualization of how to integrate social information into Ocean Accounts, drawing on the experiences of stakeholders from different social, political, cultural, and economic contexts. The paper has also provided a detailed action agenda with eight key areas for progressing the development and integration of Social Accounts in ocean accounting and decision-making. They are: (i) establish the state of evidence, (ii) develop a working Social Accounts definition, (iii) start pilot studies and use existing data and indicators, (iv) undertake gap identification, (v) learn from and connect with other disciplines, (vi) develop a standardized structure for Social Accounts, (vii) develop guidance for undertaking Social Accounts and (viii) communication and promotion of Social Accounts. We have also noted three principles which should be considered during efforts to progress Social Accounts: (i) strategic partnerships and stakeholder alignment; (ii) inclusive design and implementation; (iii) and transparency and respect for rights. We demonstrate that Social Accounts represent an emerging field of research and practice that is gaining momentum, and will soon be implemented through practical, transparent and collaborative approaches (i.e. it is not merely an aspirational goal). This paper aims to serve as a resource for relevant stakeholders and support the development of collaboration, capacity, and expertise as Pilot Social Accounts for the Ocean are developed. We hope that this paper helps to mobilize collaboration, capacity and thought leadership to develop Social Accounts for the Ocean. The perspective positions social accounting within ocean systems as a step toward advancing ecosystem-specific approaches more broadly. The 16 types of social data, while developed with marine and coastal contexts in mind, provide a conceptual foundation that can be adapted to diverse environmental contexts,

supporting the continued development of ecosystem-specific social accounting frameworks worldwide.

CRedit authorship contribution statement

R.J. Shellock: Conceptualization, Validation, Visualization, Writing – original draft, Writing – review & editing. **R.E. Thoms:** Conceptualization, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **P.A.S. James:** Funding acquisition, Supervision, Validation, Writing – original draft, Writing – review & editing. **T.G. Loureiro:** Writing – original draft, Writing – review & editing. **C. Rosado:** Writing – original draft, Writing – review & editing. **M. Swaleh:** Writing – original draft, Writing – review & editing. **E. Kurniati:** Writing – original draft, Writing – review & editing. **O. Lecuyer:** Writing – original draft, Writing – review & editing. **C. Contreras:** Writing – original draft, Writing – review & editing. **R. Arinda:** Writing – original draft, Writing – review & editing. **K.L.L. Oleson:** Writing – original draft, Writing – review & editing. **E.A. Buchary:** Writing – original draft, Writing – review & editing. **A. Rosdiana:** Writing – original draft, Writing – review & editing. **K.M. Grimsrud:** Writing – review & editing. **W. Chen:** Writing – review & editing. **G. Finau:** Writing – original draft, Writing – review & editing. **J.J. Chambo:** Writing – review & editing. **M. Granada Alarcon Blazquez:** Writing – review & editing. **N. de Wolf:** Conceptualization, Writing – review & editing. **J.A. Quijozaca Quijosaca:** Writing – review & editing. **R. Hidayat HM:** Writing – review & editing. **J. Carter:** Writing – review & editing. **S. Driscoll:** Writing – review & editing. **R. McLeod:** Writing – review & editing. **M. Ahmad:** Writing – review & editing. **A.M. Sulaiman:** Writing – review & editing. **M. Murunga:** Writing – review & editing. **A.L. Hill:** Writing – review & editing. **F. Agung:** Writing – review & editing. **M. Teoh:** Writing – review & editing. **R Cappabianca:** Writing – review & editing.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.envsci.2025.104221.

Data availability

No data was used for the research described in the article.

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